

Single choice Question with one option correct

Q.01 to Q.15

Multiple choice Question with one or more than one may be correct

Q.16 to Q.35

Numerical Based Question (no negative marking)

Q.36 to Q.55

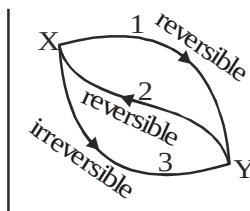
DPP Syllabus : Thermodynamic, Thermochemistry, Solid state, Chemical Kinetics

- Two first order reaction have half-lives in the ratio 8 : 1. Calculate the ratio of time intervals $t_1 : t_2$. The time t_1 and t_2 are the time period for $\left(\frac{1}{4}\right)^{\text{th}}$ completion of 1st reaction and $\left(\frac{3}{4}\right)^{\text{th}}$ completion of 2nd reaction respectively.
 (A) 1 : 0.301 (B) 0.125 : 0.602 (C) 1 : 0.62 (D) None of these
- At temperature just above 85 K, decarboxylation of acetic acid becomes a spontaneous process under standard state conditions. What is the standard entropy change (in J/K-mol) of the reaction.
 $\text{CH}_3\text{COOH}(\text{aq}) \longrightarrow \text{CH}_4(\text{g}) + \text{CO}_2(\text{g})$
Given : $\Delta H_f^\circ [\text{CH}_3\text{COOH}(\text{aq})] = -484 \text{ kJ/mole}$
 $\Delta H_f^\circ [\text{CO}_2(\text{g})] = -392 \text{ kJ/mole}$
 $\Delta H_f^\circ [\text{CH}_4(\text{g})] = -75 \text{ kJ/mole}$
 (A) 100 J/K mole (B) 200 J/K mole
 (C) 300 J/K mole (D) 400 J/K mole
- The enthalpy of formation of C_2H_6 is -84 kJ/mol . For C_3H_8 , it is -105 kJ/mol . What is the enthalpy of formation of n-pentane?
 (A) Data is insufficient (B) -147 kJ/mol
 (C) -187 kJ/mol (D) -114 kJ/mol
- What is the maximum number of layers of atoms in close packed planes that will lie within two imaginary parallel planes having a distance between them of $13 \sqrt{\frac{2}{3}} R$ in the copper crystal (FCC) ? Consider the atoms to be within the parallel planes if their centres are on or within the two parallel planes.
 [R = radius of sphere]
 (A) 5 (B) 6 (C) 7 (D) 8
- The number of atoms in 100 g of a cubic crystal containing FCC unit cell with density $d = 10 \text{ gcm}^{-3}$ and cell edge of 200 pm is equal to :
 (A) 3×10^{25} (B) 5×10^{24} (C) 1×10^{25} (D) 2×10^{25}

6. Ionic solid B^+A^- crystallizes in rock salt type of structure. 0.132 gm ionic solid salt B^+A^- is dissolved in water to make one litre solution. The pH of the solution is measured to be 6.0. If the length of face diagonal in the unit cell of B^+A^- be $600\sqrt{2}$ pm. Calculate the density of ionic solid. ($T = 298$ K), K_b for BOH is 10^{-5} . [Avogadro Number = 6.022×10^{23} , HA = strong acid]

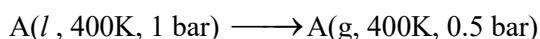
(A) 5.12 g/cc (B) 4.06 g/cc (C) 4.52 g/cc (D) 5.67 g/cc

7. Suppose a system make a transition from state X to state Y.



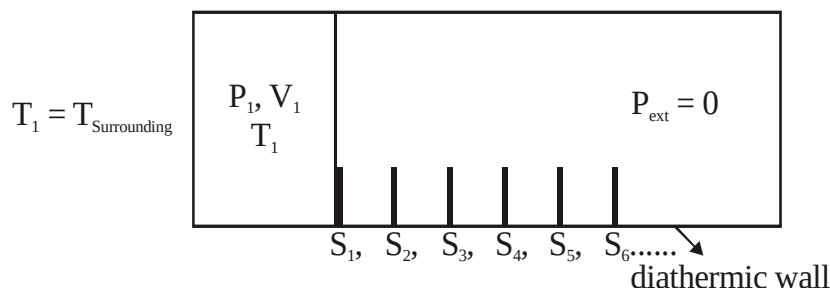
Given : $\Delta S_{XY} = 10$ J/K. Select incorrect statement.

- (A) The state Y is more disordered than state X. (B) ΔS_{XY} for path 1 and 3 is same.
 (C) $\Delta S_{YX} = -10$ J/K (D) The transition $X \rightarrow Y$ must be spontaneous.
8. The standard boiling point of A(l) is 400K. The enthalpy of vapourisation A(l) at 400 K and 1 bar is 20 kcal/mol. Which of the following information is incorrect regarding the process -



[Given : $C_{p,m}$ values (in cal / K-mol) : A(g) = 5 ; A(l) = 10; $\ln 2 = 0.7$, $\ln 5 = 1.6$, Assume ideal behaviour of A(g)]

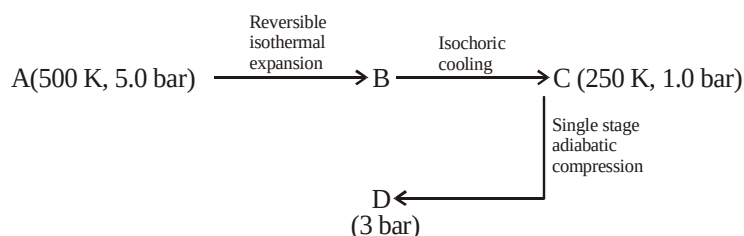
- (A) $\Delta H_m = +20$ kcal/mol (B) $\Delta S_m = +51.4$ cal/ K-mol
 (C) $\Delta G_m = -560$ cal/mol (D) The process must be spontaneous
9. Consider the process carried out for an ideal gas.



One by one, stoppers $S_1, S_2, S_3, \dots$ are removed slowly and the gas expands to V_2 L (Total volume of container) with final pressure P_2 and final temperature T_2 . Which of the following is correct for the above process.

- (A) $T_1 = T_2 \left(\frac{V_2}{V_1} \right)^{\gamma-1}$ (B) $P_1 = P_2$
 (C) $U_1 = U_2 + P_2(V_2 - V_1)$ (D) $w = 0, q = 0$

10. Choose the correct statement among the following :
- (A) Primitive cubic unit cell involve two dimensional hexagonal closed packed array of atoms
 (B) Four hexagonal closed packed two dimensional layer of atoms are involved in one FCC unit cell
 (C) Two square close packed two dimensional layer of atoms are involved in one BCC unit cell
 (D) Three square close packed two dimensional layer of atoms are involved in one BCC unit cell
11. Find type of crystallization of Ag (At. wt. = 108 amu) atom with cubical side length $5\text{\AA}$, if density of crystal lattice is 2.88 gram/cm^3 . ($N_A = 6 \times 10^{23}$)
- (A) SC (B) FCC (C) BCC (D) HCP
12. Enthalpy of disproportionation of CuCl for its one mol will be if
 $\Delta H_f^\circ[\text{CuCl}] = -100\text{ kJ/mol}$; $\Delta H_f^\circ[\text{CuCl}_2] = -400\text{ kJ/mol}$
- (A) -300 kJ/mol (B) -150 kJ/mol (C) -200 kJ/mol (D) -100 kJ/mol
13. Expansion of 1 mole of ideal gas is taking place from 2 litre to 8 litre at 300K against 1 atm pressure. Calculate ΔS_{total} in $\text{JK}^{-1}\text{mol}^{-1}$
 (given $R = \frac{8.3\text{ J}}{\text{mol-K}}$, $1\text{ lit-atm} = 100\text{ J}$, $\ln 2 = 0.693$)
- (A) 11.5 (B) 13.5 (C) 9.5 (D) 22.5
14. Reaction $A + B \longrightarrow C + D$ follow's following rate law : $\text{rate} = k = [A]^{\frac{1}{2}}[B]^{\frac{1}{2}}$. Starting with initial conc. of one mole of A and B each, what is the time taken for amount of A of become 0.25 mole. Given $k = 2.31 \times 10^{-3}\text{ sec}^{-1}$.
- (A) 300 sec. (B) 600 sec. (C) 900 sec. (D) none of these
15. For a reaction $\text{CH}_3\text{OCH}_3(\text{g}) \longrightarrow \text{CH}_4(\text{g}) + \text{H}_2(\text{g}) + \text{CO}(\text{g})$ at 750K, the rate constant is $3.52 \times 10^{-3}\text{ min}^{-1}$. Starting with a pressure of 360 mm of Hg at this temperature in a closed container, how many minutes would it take for the pressure in the container to become 760 mm Hg? (Take $\log 2.25 = 0.352$)
- (A) 50 min (B) 40.03 min (C) 23.03 min (D) 60 min
16. Two moles of an ideal gas ($C_{V,m} = \frac{3}{2}R$) is subjected to following change of state



The correct statement is/are

- (A) The pressure at B is 2.0 bar (B) The temperature at D is 450 K
 (C) $\Delta H_{CD} = 1000\text{ R}$ (D) $\Delta U_{BC} = 375\text{ R}$

17. The normal boiling point of a liquid 'A' is 350 K. ΔH_{vap} at normal boiling point is 35 kJ/mole. Pick out the correct statement(s). (Assume ΔH_{vap} to be independent of pressure).
- (A) $\Delta S_{\text{vaporisation}} > 100 \text{ J/K mole}$ at 350 K and 0.5 atm
 (B) $\Delta S_{\text{vaporisation}} < 100 \text{ J/K mole}$ at 350 K and 0.5 atm
 (C) $\Delta S_{\text{vaporisation}} < 100 \text{ J/K mole}$ at 350 K and 2 atm
 (D) $\Delta S_{\text{vaporisation}} = 100 \text{ J/K mole}$ at 350 K and 1 atm
18. Lead metal has a density of 11.1 g/cm^3 and crystallizes in a face-centred cubic lattice. Choose the correct alternatives ($\text{Pb} = 208$, $N_A = 6 \times 10^{23}$)
- (A) the volume of one unit cell is $1.25 \times 10^{-22} \text{ cm}^3$.
 (B) the volume of one unit cell is $1.25 \times 10^{-19} \text{ cm}^3$.
 (C) the atomic radius of lead is 177 pm.
 (D) the atomic radius of lead is 155 pm.
19. Which of the following statement(s) is/are correct ?
- (A) NaCl is a 'AB' crystal lattice that can be interpreted to be made up of two individual fcc unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.
 (B) In a face centred cubic unit cell, the centre of the unit cell is an octahedral void.
 (C) In fcc unit cell, octahedral and tetrahedral voids are equal in number.
 (D) Number of tetrahedral voids = $2 \times$ number of octahedral voids, is valid for ccp and hcp.
20. Which of the following is/are true ?
- (A) Ratio of nearest neighbours in simple cubic cell to next nearest neighbours in face centred cubic cell is 1.
 (B) Packing efficiency of a hypothetical unit cell in which atoms are present at each corner and each edge centre is about 26 % in metallic crystal.
 (C) Distance between two planes in FCC or HCP arrangement is same for a metal existing in both forms, with same atomic radius.
 (D) If number of unit cell along one edge of a cubic lattice are 'x', then total number of unit cell in that lattice = x^3
21. When the concentration of A is doubled, the rate for the reaction : $2A + B \rightarrow 2C$ quadruples. When the concentration of B is doubled the rate remains the same. Which mechanism below is/are not consistent with the experimental observation ?
- (A) Step I : $2A \rightarrow D$ (fast equilibrium)
 Step II : $B + D \rightarrow E$ (slow)
 Step III : $E \rightarrow 2C$ (fast)
- (B) Step I : $A + B \rightarrow D$ (fast equilibrium)
 Step II : $A + D \rightarrow 2C$ (slow)
- (C) Step I : $A + B \rightarrow D$ (slow)
 Step II : $A + D \rightarrow 2C$ (fast equilibrium)
- (D) Step I : $2A \rightarrow D$ (slow)
 Step II : $B + D \rightarrow E$ (fast)
 Step III : $E \rightarrow 2C$ (fast)

22. For a certain reaction $A \longrightarrow$ products, the $t_{1/2}$ as a function of $[A]_0$ is given as below :
- | | | | |
|------------------|---|-----|-------|
| $[A]_0$ (M) | : | 0.1 | 0.025 |
| $t_{1/2}$ (min.) | : | 100 | 50 |
- Which of the following is/are true ?
- (A) The order is $\frac{1}{2}$ (B) $t_{1/2}$ would be $100\sqrt{10}$ min for $[A]_0 = 1\text{M}$
- (C) The order is 1 (D) $t_{1/2}$ would be 100 min for $[A]_0 = 1\text{M}$
23. Which of the following statement(s) is/are **false**?
- (A) Entropy for $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$ reaction is positive
- (B) $\Delta E < 0$ for combustion of $\text{CH}_4(\text{g})$ in a sealed container with rigid adiabatic system
- (C) Change in Gibb's free energy is always zero for a reversible process in a closed system
- (D) Standard Gibb's free energy for an ideal gas reaction is a function of pressure
24. Mark the correct statement -
- (A) CsBr crystal show Schottky defect
- (B) Crystal having F-centres are coloured and paramagnetic
- (C) In Frenkel defect there is a large difference in the size of cation & anion
- (D) In Schottky defect anions are missing and electrons are present in their place
25. For the elementary reaction $A + 3B \xrightarrow{k} C$, select the correct statement -
- (A) $\frac{d[C]}{dt} = -\frac{d[A]}{dt}$
- (B) $\frac{3d[C]}{dt} = -\frac{d[B]}{dt}$
- (C) Rate law expression $r_A = k [A][B]^3$
- (D) Unit for rate of reaction is independent of order of reaction
26. Which of the following processes are spontaneous
- (A) Melting of ice at 2 atm and 273 K (B) Melting of ice at 1/2 atm and 273 K
- (C) Boiling of water at 1/2 atm and 373 K (D) Boiling of water at 2 atm and 373 K
27. For the reaction at 300K at 1 bar, $2\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \longrightarrow 2\text{C}_2(\text{g})$
- $\Delta H_f^\circ[\text{A}_2(\text{g})] = -110 \text{ cal}$
- $\Delta H_f^\circ[\text{B}_2(\text{g})] = -50 \text{ cal}$
- $\Delta H_f^\circ[\text{C}_2(\text{g})] = -400 \text{ cal}$
- Select correct statement(s), considering ideal gases -
- (A) ΔU for the reaction at 300K is positive
- (B) On increasing temperature ΔH° of reaction will increase
- (C) Reaction will be spontaneous at low temperature but not at high temperature
- (D) Reaction will be spontaneous at all temperatures

28. At 298K

$$\Delta H_{\text{formation}}^0[\text{CH}_4(\text{g})] = -75 \text{ kJ/mol}; \Delta H_{\text{formation}}^0[\text{CO}_2(\text{g})] = -395 \text{ kJ/mol}$$

$$\Delta H_{\text{combustion}}^0[\text{H}_2(\text{g})] = -285 \text{ kJ/mol}; \Delta H_{\text{combustion}}^0[\text{C}_4\text{H}_{10}(\text{g})] = -2880 \text{ kJ/mol}$$

then, which of the following is/are true -

(A) $\Delta H_{\text{formation}}^0[\text{C}_4\text{H}_{10}(\text{g})] = -125 \text{ kJ/mol}$

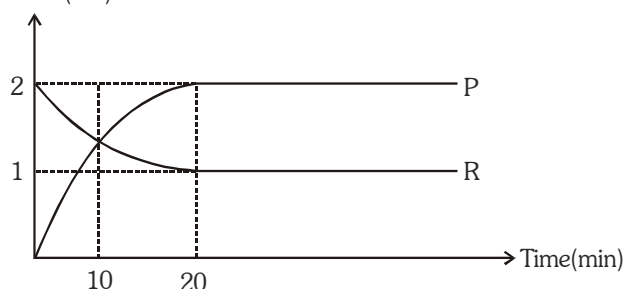
(B) $\Delta H_{\text{combustion}}^0[\text{CH}_4(\text{g})] = -890 \text{ kJ/mol}$

(C) $\Delta H_{\text{formation}}^0[\text{H}_2\text{O}(\text{l})] = -285 \text{ kJ/mol}$

(D) $\Delta H_{\text{combustion}}^0[\text{C}_{(\text{diamond})}] = -395 \text{ kJ/mol}$

29. For the reaction $\text{R}(\text{g}) \rightarrow 2\text{P}(\text{g})$

Following observation is made at 300K.

Partial
pressure (bar)

Select the correct statements (s)

(A) $(\Delta_r G)$ at 10 min = 0

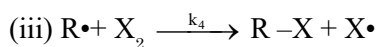
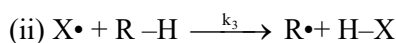
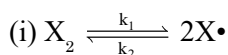
(B) $(\Delta_r G)$ at 20 min = 0

(C) $\Delta_r G^0$ at 10 min = -ve

(D) $\Delta_r G^0$ at 20 min = 0

30. For the gas phase reaction : $\text{R}-\text{H} + \text{X}_2 \rightarrow \text{R}-\text{X} + \text{HX}$

Following mechanism has been proposed :



Based on the above mechanism select the correct option (s)

(A) Effective rate constant for the formation of RX is $k_3 \sqrt{\frac{k_1}{k_2}}$

(B) $\frac{d[\text{RX}]}{dt} \propto [\text{X}_2]$

(C) Overall order of the reaction is 3/2

(D) $\frac{d[\text{RX}]}{dt} \propto [\text{RH}]$

31. For the reaction : $A \rightarrow B$. Differential rate law is : $\frac{-d[A]}{dt} = k[A]^{-1}$. If initial concentration of A is A_0 M and concentration at time 't' is A_t M then identify correct option (s)

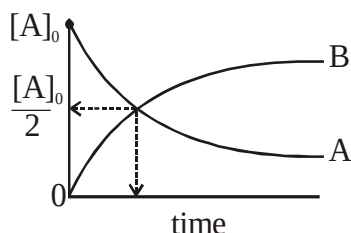
(A) $A_t^2 = A_0^2 - 2kt$

(B) Half life of A, $t_{1/2} = \frac{3A_0^2}{4K}$

(C) Completion time for reaction : $t_{100\%} = \frac{A_0^2}{2K}$

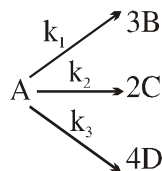
(D) The reaction must be complex

32. Which of the following is/are CORRECT :



- (A) For an elementary reaction $A \longrightarrow B$ shown in the figure the sum of slopes for curve A and B at the intersection point is zero
- (B) If the half lives of two parallel first order reactions $A \xrightarrow{\quad} B$ and $A \xrightarrow{\quad} C$ are 4 hrs and 12 hrs respectively, the average half life for decay of A is 8 hrs
- (C) In general ΔG , ΔH , ΔS all have negative values in physical adsorption at low temperature.
- (D) In CsCl type structure 8 Cs^+ ions occupy the second nearest neighbour locations of Cs^+ ion

33. For the I order parallel reactions



If $k_1 : k_2 : k_3$ is 1 : 2 : 3, then what is/are correct ($[A]_0 = 1$ M)

(A) After very long time $[C]$ is $2/3$ M

(B) $[B]_t > [C]_t$

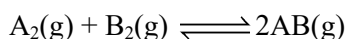
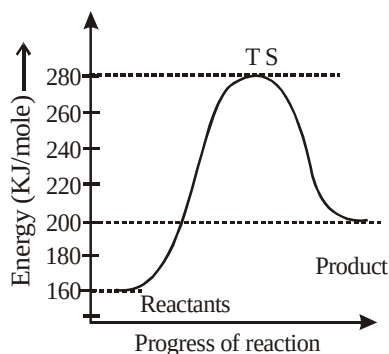
(C) $[D] = 1$ M when $[A]_t = 1/2$ M

(D) $[A]_t + \frac{[B]_t}{3} + \frac{[C]_t}{2} + \frac{[D]_t}{4} = 1\text{M}$

34. Select correct statement -

- (A) Number of second nearest atom in primitive cubic unit cell is 12.
- (B) Minimum inter-ionic distance in Na_2O is $\sqrt{3}a/2$.
- (C) Distance between two consecutive hexagonal layers of identical spherical particles of radius R in FCC is $4R\sqrt{2/3}$.
- (D) Number of next nearest neighbour atoms for S^{2-} in ZnS type structure is 12.

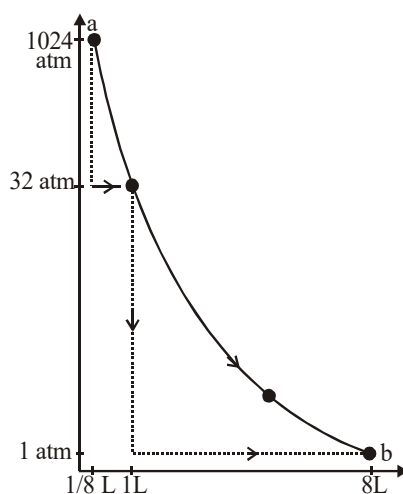
35. Consider the following hypothetical reaction and energy diagram



Which of the following is correct -

- (A) Reaction moves forward on increasing temperature
 (B) Adding He at constant pressure drives the reaction in forward direction
 (C) $\Delta H = 120 \text{ kJ/mole}$ for the above reaction
 (D) $\Delta H = 40 \text{ kJ/mole}$ for the above reaction
36. A compound formed by elements X & Y, Crystallizes in a cubic structure, where X is at the corners of the cube and Y is at six face centers. If side length is $5\text{\AA}$, estimate the density of the solid in gm/cm^3 assuming atomic weight of X and Y as 60 amu and 90 amu respectively. ($N_A = 6 \times 10^{23}$)
37. At 300 K (room temperature) a certain reaction is 50% completed in 20 minutes. At 350K the same reaction is 50% completed in 5 minutes under similar condition. Calculate the activation energy of the reaction in kcal? ($\ln 2 = 0.7$)
38. For the reaction: $3\text{A}(\text{g}) \rightarrow 2\text{B}(\text{g})$, the rate of formation of 'B' at 298 K, is represented as $\ln \left(\frac{d[\text{B}]}{dt} \right) = -0.04 + 2 \times \ln[\text{A}]$. The order of reaction is X and the value of rate constant of reaction is Y. Calculate the value of Y/X. [Given : $\ln(0.96) = -0.04$]
39. For the reaction
- $$\text{A} \longrightarrow \text{P}$$
- If ratio of $t_{7/8} : t_{1/4}$ is 7 : 2 what is order of reaction.
 Where $t_{7/8}$ = time in which 7/8 part of reactant is decayed and $t_{1/4}$ = time in which 1/4 part of reactant is decayed
40. The decomposition of a compound A in solution follow first order kinetics. If 10% solution of A is 10% decomposed in 10 minutes at 10°C then 20% solution of A is % decomposed in 20 minutes at 10°C -

41. The nonstoichiometric oxide of iron called Wustite has empirical formula Fe_xO . It has a density of 6.33 g/cm^3 , a cubic unit cell with cell constant of 400 pm and an FCC arrangement of oxygen atoms. Find the percentage of occupied iron sites ($\text{Fe} = 56$) ($N_A = 6 \times 10^{23}$)
42. The average energy required to break a P – P bond in $\text{P}_4(\text{s})$ into gaseous atoms is $53.2 \text{ kcal mol}^{-1}$. The bond dissociation energy of $\text{H}_2(\text{g})$ is $104.2 \text{ kcal mol}^{-1}$ and $\Delta_f H^\circ$ of $\text{PH}_3(\text{g})$ from $\text{P}_4(\text{s})$ is $5.1 \text{ kcal mol}^{-1}$. The P-H bond energy in kcal mol^{-1} is : [Neglect presence of van der Waals forces in $\text{P}_4(\text{s})$]
43. KCl has NaCl type face centred cubic crystal structure and CsF has CsCl type cubic crystal structure. Calculate the ratio of densities of CsF and KCl. It is given that the molar mass of CsF is twice that of KCl and edge length of KCl unit cell is 2 times that for CsF.
44. One mole of a monoatomic ideal gas is taken from a to b by expansion along two paths denoted by the solid and dashed line, as shown in the graph below. If w_A is work done along solid line path and w_B is work done along dotted line path then find $\frac{|w_A|}{|w_B|}$.



45. For the reaction
- $$\begin{array}{l} \text{A} \rightarrow 2\text{B} + \text{C} ; \Delta H = 100 \text{ kCal/mole} \\ \text{A} \rightarrow \text{D} + \text{E} ; \Delta H = 200 \text{ kCal/mole} \end{array}$$

If initially 1 mole of A is taken & finally ratio of products B & D is 6 : 1, calculate the enthalpy change (in kCal) per mole of A reacted.

46. Find bond enthalpy of C = O (in kJ/mol) using following informations

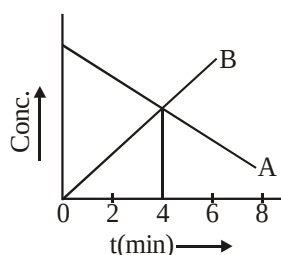
$$\Delta H_{\text{atomisation}} [\text{C(s)}] = 700 \text{ kJ/mol}$$

$$\Delta H_f[\text{CO}_2(\text{g})] = -400 \text{ kJ/mol}$$

$$\text{BE}_{\text{O}=\text{O}} = 500 \text{ kJ/mol}$$

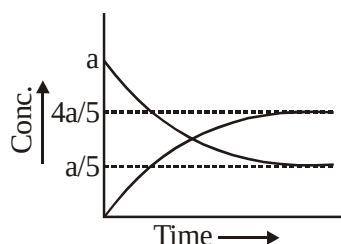
$$\text{Resonance energy of CO}_2 = -150 \text{ kJ/mol}$$

47. A cubic unit cell contains ' A^{x+} ' ions at the corners and ' B^{y-} ' ions at the centre of each face.
- (i) The difference of coordination number of A and B = p (Assuming no anion-anion or no cation-cation contact)
- (ii) If all atoms along one of the C_4 axis are removed, then the formula becomes A_qB_r . Find $\frac{p \times q}{r}$.
48. For zero order reaction $\text{A} \rightarrow 3\text{B}$ plot of concentration vs time is as follows.



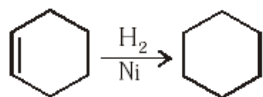
Find half life period in minutes if the initial concentration of A be 1 M.

49. A reversible reaction which is 1st order in both direction is $\text{A} \xrightleftharpoons[K_2]{K_1} \text{B}$. For which value of $\frac{K_1}{K_2}$ following graph will be correct (where a is initial concentration of reactant A)

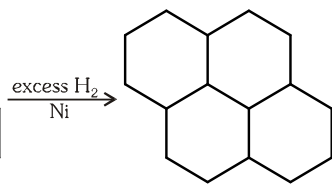
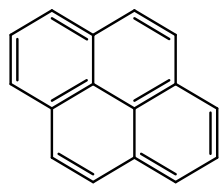


50. In first order reaction ($\text{A} \rightarrow \text{P}$) calculate $\frac{t}{t_{1/2}}$ if $\frac{15}{16}$ fraction of reactant is decayed in 't' time where $t_{1/2}$ is half life.

51. Use the following data to answer the question below :

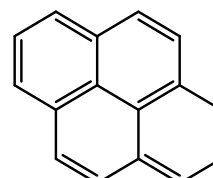


$$\Delta H = -30 \text{ kcal mol}^{-1};$$

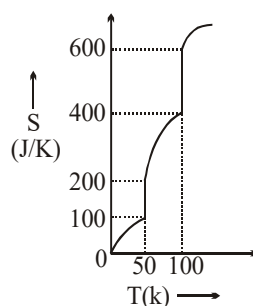


$$\Delta H = -180 \text{ kcal mol}^{-1}$$

Calculate the magnitude of resonance energy (in kcal/mole) for



52. For 1 mole substance variation of absolute entropy with temperature is given by following graph. What is molar enthalpy of vapourisation (ΔH_{vap} in kJ) of substance.



53. For a reaction : $2\text{A(s)} + \text{B(g)} \longrightarrow 3\text{C(l)}$
 standard entropy change of reaction is 2 kJ/mol-K and standard enthalpy of combustion of A, B and C are -100 , -60 -285 kJ/mol respectively, then find the magnitude of maximum useful work that can be obtained at 27°C and 1 bar pressure for reaction (in kJ/mol).
54. What is the $S^\circ(\text{OH}^-)$ (in cal/mol K) at 300 K
 Given : $K_w(\text{H}_2\text{O}) = 10^{-14}$
 $\Delta H_{\text{neut}}(\text{H}^+ + \text{OH}^-) = -13.5 \text{ kcal.}$
 $S^\circ_{\text{H}_2\text{O}} = 16 \text{ cal/mol K}$; $R = 2 \text{ cal/mole K}$; $\ln x = 2 \log x$ (Take : $S^\circ_{\text{H}^+} = 0$ at 300K)
55. Calculate partial pressure of HCl (g) in bar at TK above the sample of $\text{NH}_4\text{Cl(s)}$ as result of its decomposition.
 Given data : $\Delta G_f^\circ [\text{NH}_4\text{Cl (s)}] = -100 \text{ kJ mol}^{-1}$
 $\Delta G_f^\circ [\text{NH}_3 \text{ (g)}] = -17 \text{ kJ mol}^{-1}$
 $\Delta G_f^\circ [\text{HCl (g)}] = -97 \text{ kJ mol}^{-1}$
 Use : $RT = 10000 \text{ J mole}^{-1}$ & $e^{0.7} = 2$

ANSWER-KEY DPP # 07 (MAIN TO ADVANCED)

1. (C)	2. (B)	3. (B)	4. (C)	5. (B)
6. (B)	7. (D)	8. (D)	9. (D)	10. (B)
11. (C)	12. (D)	13. (C)	14. (B)	15. (C)
16. (ABC)	17. (ACD)	18. (AC)	19. (ABD)	20. (ABCD)
21. (ABC)	22. (AB)	23. (BCD)	24. (ABC)	25. (ABCD)
26. (AC)	27. (AC)	28. (ABC)	29. (BC)	30. (ACD)
31. (ACD)	32. (AC)	33. (ACD)	34. (AD)	35. (AD)
36. (4.40 g/cm ³)	37. (5.88)	38. (0.24)	39. (0)	40. (19)
41. (80)	42. (77)	43. (4)	44. (5.14)	45. (125)
46. (725)	47. (4)	48. (8)	49. (4)	50. (4)
51. (60)	52. (20)	53. (5)	54. (5)	55. (2)